

Untitled3

May 19, 2026

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[2]: import pandas as pd
import numpy as np
import statsmodels.api as sm
import statsmodels.formula.api as smf

# 1. Load the dataset directly from the UCLA server
url = "https://stats.idre.ucla.edu/stat/data/binary.csv"
df = pd.read_csv(url)

# 2. Treat 'rank' as a categorical factor variable (Tiers 1 to 4)
df['rank'] = df['rank'].astype('category')

# 3. Fit the Logistic Regression Model using GLM (Binomial family with Logit link)
model = smf.glm(formula="admit ~ gre + gpa + C(rank)", data=df, family=sm.families.Binomial()).fit()

# 4. Display the comprehensive model summary
print("==== LOGISTIC REGRESSION MODEL SUMMARY =====")
print(model.summary())

# 5. Calculate Odds Ratios (OR) and their corresponding 95% Confidence Intervals
print("\n===== CALCULATED ODDS RATIOS AND 95% CI =====")
params = model.params
conf = model.conf_int()
odds_ratios = np.exp(params)
conf_odds = np.exp(conf)

# Combine coefficients, odds ratios, and p-values into a clean DataFrame
results_df = pd.DataFrame({
    'Coefficient (Beta)': params,
    'Odds Ratio (e^Beta)': odds_ratios,
    'Lower CI (Odds)': conf_odds[0],
    'Upper CI (Odds)': conf_odds[1],
    'p-value': model.pvalues
})
```

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print(results_df.round(4))
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===== LOGISTIC REGRESSION MODEL SUMMARY =====

Generalized Linear Model Regression Results

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Dep. Variable:          admit    No. Observations:          400
Model:                GLM      Df Residuals:              394
Model Family:         Binomial Df Model:                  5
Link Function:         Logit   Scale:                   1.0000
Method:                IRLS    Log-Likelihood:        -229.26
Date:                  Tue, 19 May 2026 Deviance:           458.52
Time:                  00:08:56 Pearson chi2:         397.
No. Iterations:        4       Pseudo R-squ. (CS):      0.09846
Covariance Type:       nonrobust
=====
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	coef	std err	z	P> z	[0.025	0.975]
Intercept	-3.9900	1.140	-3.500	0.000	-6.224	-1.756
C(rank)[T.2]	-0.6754	0.316	-2.134	0.033	-1.296	-0.055
C(rank)[T.3]	-1.3402	0.345	-3.881	0.000	-2.017	-0.663
C(rank)[T.4]	-1.5515	0.418	-3.713	0.000	-2.370	-0.733
gre	0.0023	0.001	2.070	0.038	0.000	0.004
gpa	0.8040	0.332	2.423	0.015	0.154	1.454

===== CALCULATED ODDS RATIOS AND 95% CI =====

	Coefficient (Beta)	Odds Ratio (e^Beta)	Lower CI (Odds)	\
Intercept	-3.9900	0.0185	0.0020	
C(rank)[T.2]	-0.6754	0.5089	0.2737	
C(rank)[T.3]	-1.3402	0.2618	0.1331	
C(rank)[T.4]	-1.5515	0.2119	0.0934	
gre	0.0023	1.0023	1.0001	
gpa	0.8040	2.2345	1.1661	

	Upper CI (Odds)	p-value
Intercept	0.1728	0.0005
C(rank)[T.2]	0.9464	0.0328
C(rank)[T.3]	0.5151	0.0001
C(rank)[T.4]	0.4807	0.0002
gre	1.0044	0.0385
gpa	4.2819	0.0154

Problem 5: Logistic Regression Analysis (Binary GLM) for Graduate School Admission

This section evaluates the impact of academic predictors—Graduate Record Examination (GRE) scores, Cumulative Grade Point Average (GPA), and the prestige/rank of the applicant’s undergraduate institution (ranging from Rank 1 as highest prestige to Rank 4 as lowest)—on the probability of graduate school admission. Since the response variable (`admit`) is binary (1 = Admitted, 0 = Not Admitted), a Generalized Linear Model (GLM) using the Binomial family with a logit link function was constructed.

1. Hypothesis Formulation

H₀: Academic predictors (GRE, GPA) and undergraduate institutional prestige (Rank) have no statistically significant effect on the probability of graduate school admission.

H₁: At least one of the predictors (GRE, GPA, or institutional Rank) significantly affects the probability of graduate school admission.

2. Model Specification

The structural logistic regression equation is defined as follows:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1(\text{C(rank)[T.2]}) + \beta_2(\text{C(rank)[T.3]}) + \beta_3(\text{C(rank)[T.4]}) + \beta_4(\text{gre}) + \beta_5(\text{gpa})$$

Where $p = P(\text{admit} = 1)$, and undergraduate institutional Rank 1 is treated as the reference baseline category.

3. Empirical Model Estimation Results

The maximum likelihood estimates, associated standard errors, asymptotic Wald z -metrics, Wald p -values, and calculated Odds Ratios (OR) along with their 95% confidence intervals are structured from the model output in the table below:

Predictor Variable	Coef (β)	Std Error	z -value	p -value	Odds Ratio	95% CI (Odds)
Intercept	-3.9900	1.1400	-3.500	0.0000	0.0185	[0.0020, 0.1728]
C(rank)[T.2]	-0.6754	0.3160	-2.134	0.0330	0.5089	[0.2737, 0.9464]
C(rank)[T.3]	-1.3402	0.3450	-3.881	0.0000	0.2618	[0.1331, 0.5151]
C(rank)[T.4]	-1.5515	0.4180	-3.713	0.0000	0.2119	[0.0093, 0.4807]
gre	0.0023	0.0010	2.070	0.0380	1.0023	[1.0001, 1.0044]
gpa	0.8040	0.3320	2.423	0.0150	2.2345	[1.1661, 4.2819]

4. Statistical Interpretation and Findings

The comprehensive Generalized Linear Model parameters reveal that academic qualifications and institutional reputation both serve as critical, statistically significant drivers of graduate school admission thresholds at the nominal level ($\alpha = 0.05$):

- **GRE Scores** ($p = 0.0385$): Holding GPA and institutional rank constant, a one-unit increase in an applicant's GRE score significantly increases the log-odds of admission by 0.0023. This corresponds to an Odds Ratio of 1.0023 (95% CI: [1.0001, 1.0044]), indicating a minor but steady positive effect for every incremental point scored.
- **GPA** ($p = 0.0154$): Academic performance in the form of cumulative GPA yields a highly potent and statistically significant positive impact. For every 1-point increase in GPA, the log-odds of being admitted increase by 0.8040. The calculated Odds Ratio ($OR = 2.2345$, 95% CI: [1.1661, 4.2819]) reveals that an applicant's odds of admission more than double for each full unit increase in cumulative GPA.
- **Institutional Prestige/Rank**: The categorical institutional tier parameters are evaluated against the highly prestigious baseline (Rank 1). All lower tiers exhibit a stark, negative coefficient that is highly significant:
 - Applicants from **Rank 2 institutions** experience a significant drop in admission odds ($OR = 0.5089$), representing roughly a 49.11% reduction in odds compared to Rank 1.
 - Applicants from **Rank 3 institutions** experience an even deeper disadvantage ($OR = 0.2618$), meaning their odds of admission are reduced by approximately 73.82% compared to the baseline.
 - Applicants from the least prestigious **Rank 4 institutions** demonstrate the lowest likelihood of success ($OR = 0.2119$), experiencing a massive 78.81% drop in comparison to the Rank 1 baseline.

In summary, higher numeric scores (GRE and GPA) significantly boost an applicant's chances of admission, whereas graduating from a lower-prestige tier undergraduate institution substantially dampens the empirical likelihood of securing graduate school enrollment.